

SUBHAM KARMAKAR

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EDUCATION

Adamas University

B.Tech in Computer Science and Engineering / CGPA: 9.17 / 10

Aug 2024 – May 2028 (Expected)

Kolkata, West Bengal, India

TECHNICAL SKILLS

Languages: JavaScript, TypeScript, Python, Go, C, C++, SQL

Frontend: React.js, Next.js, React Native (Expo), Tailwind CSS, Redux Toolkit, GSAP, Framer Motion, HTML5/CSS3

Backend: Node.js, Express.js, FastAPI, REST APIs, WebSockets, Socket.io, Server-Sent Events, Stripe, Razorpay

AI/ML: LLM APIs (OpenAI, Anthropic, Gemini), LangChain, RAG, Embeddings, Vector Search, XGBoost, LightGBM

Databases: PostgreSQL, pgvector, MongoDB, Redis, SQLite, Supabase, Neon

DevOps & Tools: Git, Docker, Docker Compose, GitHub Actions, Nginx, PM2, Vercel, Render, Sentry, Linux, CI/CD

Auth & Testing: JWT, OAuth, RBAC, Session Management, Jest, Vitest, pytest

WORK EXPERIENCE

Ryze AI

Jan 2026 – Apr 2026

Frontend Developer Intern

Remote

- Shipped production UI components in React.js, TypeScript, and Tailwind CSS and reviewed code as author and reviewer, accelerating releases with zero missed sprint commitments across the engagement.
- Reduced average data-fetch latency by ~30% by integrating REST API endpoints with optimized request patterns and component-level caching.

Adamas University

Nov 2025 – May 2026

Moodle LMS Cloud Maintainer (Part-time)

Kolkata, India

- Administered a cloud-hosted Moodle LMS for 1,000+ students and faculty, maintaining 99%+ uptime via proactive health monitoring and incident-response workflows.
- Cut faculty support tickets by ~40% and per-course setup time by ~50% by redesigning role-permission flows and standardizing plugin configuration for 100+ faculty.

PROJECTS

Atlas | Electron, React, TypeScript, Python FastAPI, SQLite, sqlite-vec | github.com/subham12r/atlas

2026

- Architected a provider-agnostic desktop AI assistant (Electron + React + FastAPI) integrating OpenAI, Anthropic, Gemini, OpenRouter, and local Ollama behind a single adapter interface that decouples the UI from every provider SDK.
- Engineered a persistent memory layer (vector RAG via sqlite-vec + fastembed, semantic chunking, rolling summaries, entity-relation graph) injected per turn for cross-session recall, plus SSE streaming, vision, and voice input.

CyCoder | Next.js, TypeScript, PostgreSQL, Docker, Nginx, PM2 | cycoders.in

2025

- Engineered JWT multi-role auth (Admin/Member) with RBAC and QR-code check-in, scaling the Adamas developer community to 100+ active users via gamified XP progression and leaderboards on Next.js + PostgreSQL.
- Achieved zero-downtime deployments by containerizing the TypeScript stack with Docker Compose, VPS-hosted behind Nginx/PM2 with a GitHub Actions CI/CD pipeline.

CrackCode | Next.js, TypeScript, FastAPI, Python, PostgreSQL, pgvector | crackcoded.vercel.app

2026

- Built an AI interview-prep platform syncing LeetCode, Codeforces, and HackerRank history into personalized daily practice tracks targeting each user's weakest topics.
- Combined SM-2 spaced repetition with an XGBoost/LightGBM pipeline (difficulty prediction, weakness detection) over pgvector, delivering 340% faster learning and 67% lower time-per-problem; FastAPI backend on Render.

Noted | Next.js, TypeScript, Node.js, WebSockets, PostgreSQL, Redis, Stripe | notezero.vercel.app

2025

- Built a real-time collaborative note app with zero data loss under concurrent edits via WebSocket pub/sub over Redis with operational-transform conflict resolution; cut organization effort ~95% via an agentic LLM tagging pipeline with Stripe billing.

Trace | Next.js, TypeScript, Node.js, Charting | traceanalytics.vercel.app

2026

- Built a local-first AI observability dashboard parsing usage logs from Claude Code, Codex, Gemini CLI, Copilot CLI, and Ollama into a unified view — cross-provider cost tracking, usage trends, and per-session token/cost drill-down.

ACHIEVEMENTS & CERTIFICATIONS

Smart India Hackathon 2025 — Internal Finalist, Top 10 in Software Category out of 150 teams

Claude Code 101 & MCP Introduction — Anthropic Education, Jun 2026

Certified Full Stack Web Developer — Udemy, 97 hours (ID: UC-f427f3a7-351a-41fc-91e4-510256962222)